

Cutaneous T-cell Lymphoma (CTCL) and Extracorporeal Photopheresis (ECP)

Which patients should be considered for ECP treatment?

Extracorporeal photopheresis (ECP) should be considered for patients with **cutaneous T-cell lymphoma (CTCL)**, particularly in the following settings:

- **Sézary syndrome (SS):** ECP is a first-line systemic therapy for patients with leukemic involvement, given its ability to target circulating malignant T-cells.
- **Advanced-stage mycosis fungoides (MF):** Patients with stage IIB–IV disease, especially those with blood involvement (e.g., B1 or B2 according to the ISCL/EORTC staging) or extensive skin burden (e.g., erythroderma), are strong candidates for ECP.
- **Early-stage MF (IB–IIA) with refractory or progressive disease:** ECP may be used when skin-directed therapies have failed or are not feasible.
- **Patients with erythroderma:** ECP is particularly effective in managing erythrodermic CTCL, regardless of Sézary cell count.
- **Combination therapy settings:** ECP is often combined with other systemic agents (e.g., retinoids like bexarotene, interferon-alpha, or monoclonal antibodies such as mogamulizumab) to enhance efficacy.

ECP is especially beneficial for patients who require long-term systemic therapy but need a treatment with a favorable safety profile and minimal immunosuppression.

What is the optimal treatment schedule, and how long should ECP treatment be continued?

- **Initial induction phase:**

ECP is typically administered **twice weekly** (e.g., on two consecutive days every 2–3 weeks) for **6–12 months**, depending on clinical response.

- **Maintenance phase:**

After achieving a response (partial or complete), treatment frequency is gradually reduced based on clinical improvement and disease stability.

Common regimens include: - Once every 2 weeks - Once monthly

Maintenance therapy is generally continued for **1–2 years or longer** in responders.

- **Duration of therapy:**

Treatment should be continued as long as there is **evidence of clinical benefit** and no disease progression. Some patients remain on long-term ECP for several years, especially those with chronic, relapsing disease.

- **Treatment adjustments:**

Frequency may be increased during disease flares or reduced in sustained remission. Individualization based on response, tolerability, and disease burden is key.

How can therapeutic efficacy be assessed?

Therapeutic efficacy of ECP in CTCL should be evaluated using standardized response criteria, such as those defined by the **International Society for Cutaneous Lymphomas (ISCL)** and **European Organisation for Research and Treatment of Cancer (EORTC)**:

- **Skin assessment:**

Use of the **modified Severity-Weighted Assessment Tool (mSWAT)** to quantify skin lesion burden, including patches, plaques, and tumors.

- **Blood involvement:**

Evaluation of peripheral blood for Sézary cells using:

- Absolute Sézary cell count
- Flow cytometry for CD4+/CD7– or CD4+/CD26– phenotypes
- T-cell receptor gene rearrangement (clonality)
- **Lymph nodes and visceral organs:**
Imaging (e.g., CT, PET-CT) and biopsies if indicated to assess nodal or internal organ involvement.
- **Response criteria:**
 - **Complete response (CR):** Disappearance of all clinical, histologic, and laboratory signs of disease.
 - **Partial response (PR):** $\geq 50\%$ improvement in skin score (mSWAT), with improvement in other involved sites and no new lesions.
 - **Stable disease (SD):** Less than PR but no progression.
 - **Progressive disease (PD):** $\geq 25\%$ increase in skin score or new lesions or worsening of blood/organ involvement.
- **Symptom assessment:**
Monitoring of **pruritus** (e.g., visual analog scale), quality of life (e.g., Skindex-16), and erythroderma.
- **Time to response:**
Responses may take **3–6 months** to become evident; therefore, treatment should generally continue for at least 6 months before assessing lack of efficacy.

Regular follow-up every 3 months during therapy is recommended to monitor response and adjust treatment accordingly.